Notebook 5 — Chapter 5.3

Geometric Probability & 2D Integrals

20 worked examples — uniform distributions on 2D regions, sums and absolute differences of independent uniforms, and stick-breaking puzzles — calibrated for Stage 1/Stage 2 speed and rigor

The one master idea. If (X, Y) is uniform on a region S, then for any event A

$$\mathbb{P}\big((X,Y)\in A\big) = \frac{\text{Area}(A\cap S)}{\text{Area}(S)}$$

Every example below is: *draw the region, shade the event, compute an area* (by geometry when the boundary is straight, by an integral when it curves).

A. Uniform Distributions on 2D Regions

Example 1 — Probability = Area Ratio on a Rectangle

(X, Y) is uniform on $[0, 2] \times [0, 3]$. Find $\mathbb{P}(X + Y \le 2)$.
Solution. The density is constant: $f = 1/\text{Area} = 1/6$. The event $x+y \le 2$ is the triangle with vertices $(0, 0), (2, 0), (0, 2)$. It sits entirely inside the rectangle (since $y \le 2 < 3$ and $x \le 2$), so no clipping is needed.

$$\text{Area} = \tfrac{1}{2}(2)(2) = 2$$

$$\mathbb{P}(X+Y\le 2) = \frac{2}{6} = \boxed{\tfrac{1}{3}}$$

⋆ *Analogy:* a dart thrown blindly at a rug. The chance of hitting a patch is just the patch's share of the rug's area — no calculus, just a ruler. The *only* skill is checking whether the patch spills outside the rug (then you must clip it).

Example 2 — Uniform on the Unit Disk: the Radial Law

(X, Y) is uniform on $x^2 + y^2 \le 1$. Let $R = \sqrt{X^2 + Y^2}$. Find (a) the law of R and $\mathbb{E}[R]$, (b) $\mathbb{P}(R < \frac{1}{2})$, (c) $\mathbb{P}(|X| + |Y| \le 1)$.
Solution. (a) $\{R \le r\}$ is a smaller disk of area πr^2:

$$\mathbb{P}(R\le r) = \frac{\pi r^2}{\pi} = r^2 \;\Rightarrow\; f_R(r) = 2r, \quad 0\le r\le 1$$

$$\mathbb{E}[R] = \int_0^1 r\cdot 2r\,dr = \boxed{\tfrac{2}{3}}$$

(b) $\mathbb{P}(R < \frac{1}{2}) = (\frac{1}{2})^2 = \boxed{\frac{1}{4}}$.
(c) $|x| + |y| \le 1$ is a diamond with diagonals of length 2, so area $= \frac{1}{2}(2)(2) = 2$. It lies inside the disk, hence

$$\mathbb{P} = \frac{2}{\pi} = \boxed{\approx 0.637}$$

⋆ *Analogy:* rings of a dartboard. The ring at radius r has circumference $2\pi r$ — it grows with r — so a uniform dart is far more likely to land in the outer rings. That is why $\mathbb{E}[R] = \frac{2}{3}$ and not $\frac{1}{2}$: *uniform in area is not uniform in radius.*

Example 3 — A Marginal of the Disk Is Not Uniform

Same uniform disk (density $\frac{1}{\pi}$). Find $f_X(x)$ and $\mathbb{P}(|X| \le \frac{1}{2})$.
Solution. For fixed x, y runs over $|y| \le \sqrt{1-x^2}$:

$$f_X(x) = \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \frac{1}{\pi}\,dy = \frac{2}{\pi}\sqrt{1-x^2}$$

Integrate with the antiderivative $\int\sqrt{1-x^2}\,dx = \frac{1}{2}\left(x\sqrt{1-x^2} + \arcsin x\right)$:

$$\mathbb{P}(|X|\le\tfrac{1}{2}) = \frac{2}{\pi}\cdot 2\cdot\tfrac{1}{2}\left(\tfrac{1}{2}\cdot\tfrac{\sqrt{3}}{2} + \tfrac{\pi}{6}\right)$$

$$= \frac{\sqrt{3}}{2\pi} + \frac{1}{3} = \boxed{\approx 0.609}$$

▶ *Speed trick:* a uniform joint law does *not* give uniform marginals unless the region is a rectangle. The marginal at x is proportional to the *height of the slice* of the region above x — here a semicircle.

Example 4 — Uniform on a Triangle: Marginals and Covariance

(X, Y) is uniform on $T = \{0 < y < x < 1\}$. Find the marginals, $\mathbb{E}[X], \mathbb{E}[Y]$ and $\text{Cov}(X, Y)$.

Solution. $\text{Area}(T) = \frac{1}{2}$, so the density is $f = 2$ on T.

$$f_X(x) = \int_0^x 2\,dy = 2x, \qquad f_Y(y) = \int_y^1 2\,dx = 2(1-y)$$

$$\mathbb{E}[X] = \int_0^1 2x^2\,dx = \tfrac{2}{3}, \qquad \mathbb{E}[Y] = \int_0^1 2y(1-y)\,dy = \tfrac{1}{3}$$

$$\mathbb{E}[XY] = \int_0^1\int_0^x 2xy\,dy\,dx = \int_0^1 x^3\,dx = \tfrac{1}{4}$$

$$\text{Cov}(X,Y) = \tfrac{1}{4} - \tfrac{2}{3}\cdot\tfrac{1}{3} = \boxed{\tfrac{1}{36}}$$

⋆ *Analogy:* the support is a ramp: Y can never exceed X. Knowing X caps how high Y can go, so the two are dependent (and positively correlated) even though the density is flat — the *shape of the region* is the coupling.

Example 5 — Distance to the Nearest Side of a Square

(X, Y) is uniform on $[0, 1]^2$. Let $D = \min(X, 1 - X, Y, 1 - Y)$. Find $\mathbb{E}[D]$ and $\mathbb{P}(D > 0.1)$.
Solution. $D > d$ means the point is at least d from every side, i.e. it lies in the shrunken square $[d, 1-d]^2$, of side $1 - 2d$:

$$\mathbb{P}(D>d) = (1-2d)^2, \qquad 0\le d\le\tfrac{1}{2}$$

$$\mathbb{P}(D>0.1) = 0.8^2 = \boxed{0.64}$$

Use $\mathbb{E}[D] = \int_0^{1/2}\mathbb{P}(D>d)\,dd$:

$$\mathbb{E}[D] = \int_0^{1/2}(1-2d)^2 dd = \Big[-\tfrac{(1-2d)^3}{6}\Big]_0^{1/2} = \boxed{\tfrac{1}{6}}$$

⋆ *Analogy:* peeling a picture frame. Each time you strip a border of width d off a square rug, the "safe interior" shrinks to side $1 - 2d$; the area left over is your survival probability.

B. Sums and Absolute Differences of Independent Uniforms

Example 6 — The Law of $X + Y$ (the Tent)

X, Y iid Uniform$(0, 1)$, $S = X + Y$. Find the CDF of S and $\mathbb{P}(S > 1.5)$.
Solution. (X, Y) is uniform on the unit square, so $\mathbb{P}(S \le s)$ is the area under the line $x + y = s$.
Case $0 \le s \le 1$*:* a right triangle with legs s: $\mathbb{P}(S\le s) = \frac{s^2}{2}$.
Case $1 \le s \le 2$*:* the complement is a corner triangle with legs $2 - s$:

$$\mathbb{P}(S\le s) = 1 - \tfrac{(2-s)^2}{2}$$

Differentiating gives the density $f_S(s) = s$ on $[0, 1]$ and $2 - s$ on $[1, 2]$ — a tent peaking at $s = 1$.

$$\mathbb{P}(S>1.5) = \frac{(2-1.5)^2}{2} = \boxed{\tfrac{1}{8}}$$

⋆ *Analogy:* two dice. There is exactly one way to roll a 2 but six ways to roll a 7, so sums pile up in the middle. The continuous version of "more ways to make the middle" is the length of the diagonal slice $x + y = s$, which is longest at $s = 1$.

Example 7 — The Law of $|X - Y|$

X, Y iid Uniform$(0, 1)$, $D = |X - Y|$. Find the law of D, $\mathbb{E}[D]$, $\text{Var}(D)$ and $\mathbb{P}(D \le 0.2)$.
Solution. $\{D > d\} = \{X - Y > d\} \cup \{Y - X > d\}$: two corner triangles, each with legs $1 - d$:

$$\mathbb{P}(D>d) = 2\cdot\tfrac{(1-d)^2}{2} = (1-d)^2$$

$$\mathbb{P}(D\le d) = 2d - d^2, \qquad f_D(d) = 2(1-d)$$

$$\mathbb{E}[D] = \int_0^1 (1-d)^2 dd = \boxed{\tfrac{1}{3}}$$

$$\mathbb{E}[D^2] = \int_0^1 d^2\cdot 2(1-d)\,dd = \tfrac{1}{6} \;\Rightarrow\; \text{Var}(D) = \tfrac{1}{6} - \tfrac{1}{9} = \boxed{\tfrac{1}{18}}$$

$$\mathbb{P}(D\le 0.2) = 0.4 - 0.04 = \boxed{0.36}$$

⋆ *Analogy:* drop two beads on a unit string. The typical gap between them is $\frac{1}{3}$ — and the density $2(1-d)$ says small gaps are more common than large ones.

Example 8 — The Meeting Problem

Two traders arrive independently and uniformly during a 60-minute window. Each waits 15 minutes for the other, then leaves. Find $\mathbb{P}$(they meet).
Solution. With arrival times $X, Y \sim \text{Unif}(0, 60)$, they meet iff $|X-Y| \le 15$. Normalize by 60: this is $|U-V| \le \frac{1}{4}$ on the unit square. By Example 7,

$$\mathbb{P}(|U-V| \le \tfrac{1}{4}) = 1 - \left(1-\tfrac{1}{4}\right)^2 = 1 - \tfrac{9}{16} = \boxed{\tfrac{7}{16}}$$

Geometrically: the "miss" region is two corner triangles with legs 45 inside a 60×60 square, so $\mathbb{P}(\text{miss}) = \frac{45^2}{60^2} = \frac{9}{16}$.
▶ *Speed trick:* for $|X-Y| \le w$ on $[0,1]^2$ the answer is $1-(1-w)^2 = 2w - w^2$ — memorize it, and just rescale time to a unit interval first.

Example 9 — Unequal Intervals: Clip the Triangle

$X \sim \text{Unif}(0,2)$ and $Y \sim \text{Unif}(0,3)$ are independent. Find (a) $\mathbb{P}(X+Y \le 4)$ and (b) $\mathbb{P}(Y > X)$.
Solution. The pair is uniform on a rectangle of area 6.
(a) The complement $x + y > 4$ is the corner triangle with vertices $(1,3), (2,3), (2,2)$ (the line $x+y=4$ meets $y=3$ at $x=1$ and $x=2$ at $y=2$). Its legs are 1 and 1:

$$\mathbb{P}(X+Y>4) = \frac{\frac{1}{2}}{6} = \frac{1}{12} \;\Rightarrow\; \mathbb{P}(X+Y \le 4) = \boxed{\tfrac{11}{12}}$$

(b) The complement $y \le x$ is the triangle $(0,0), (2,0), (2,2)$, of area 2:

$$\mathbb{P}(Y > X) = 1 - \frac{2}{6} = \boxed{\tfrac{2}{3}}$$

⋆ *Analogy:* the formula "triangle over rectangle" only works if the cutting line stays inside the rectangle. Always find where the line hits the *edges* of the rectangle first; that tells you which corner (or strip) you are really measuring.

Example 10 — A Curved Boundary: $\mathbb{P}(XY \le \frac{1}{2})$

X, Y iid Uniform$(0,1)$. Find $\mathbb{P}(XY \le \frac{1}{2})$.
Solution. The boundary $xy = \frac{1}{2}$ is a hyperbola, so we integrate. Compute the complement $\{xy > \frac{1}{2}\}$: it needs $x > \frac{1}{2}$, and then $y \in \left(\frac{1}{2x}, 1\right)$:

$$\text{Area} = \int_{1/2}^{1} \left(1 - \frac{1}{2x}\right) dx = \tfrac{1}{2} - \tfrac{1}{2}\ln 2$$

$$\mathbb{P}(XY \le \tfrac{1}{2}) = 1 - \left(\tfrac{1}{2} - \tfrac{\ln 2}{2}\right) = \boxed{\tfrac{1}{2} + \tfrac{\ln 2}{2} \approx 0.8466}$$

▶ *Speed trick:* for a product or ratio the natural cut is the hyperbola/line through the origin, so integrate along the variable where the boundary is a function of one variable and watch the limits (here x must exceed $\frac{1}{2}$, otherwise $y \le 1$ cannot reach $xy > \frac{1}{2}$).

C. Stick-Breaking Puzzles

Example 11 — Why the Triangle Probability Is $\frac{1}{4}$

A unit stick is broken at two independent Uniform$(0,1)$ points. Find $\mathbb{P}$(the 3 pieces form a triangle).
Solution. Sort the cuts: $A < B$. The pair (A, B) is uniform on $\{0 < a < b < 1\}$, of area $\frac{1}{2}$. The pieces are $a,\ b-a,\ 1-b$.
The triangle inequality (each piece shorter than the sum of the other two) is equivalent to *every piece* $< \frac{1}{2}$:

$$a < \tfrac{1}{2}, \qquad b > \tfrac{1}{2}, \qquad b - a < \tfrac{1}{2}$$

In the (a,b)-plane this is the triangle with vertices $(0, \frac{1}{2}), (\frac{1}{2}, \frac{1}{2}), (\frac{1}{2}, 1)$, with legs $\frac{1}{2}$ and $\frac{1}{2}$:

$$\text{Area} = \tfrac{1}{2} \cdot \tfrac{1}{2} \cdot \tfrac{1}{2} = \tfrac{1}{8}$$

$$\mathbb{P}(\text{triangle}) = \frac{1/8}{1/2} = \boxed{\tfrac{1}{4}}$$

⋆ *Analogy:* a piece is a "bully" if it is $\ge$ half the stick: the other two together cannot reach across. A triangle exists exactly when there is no bully.

Example 12 — The Simplex View: $\mathbb{P}(\text{piece} > t) = (1-t)^2$

Same stick, pieces (a,b,c). Show $\mathbb{P}(a > t) = (1-t)^2$ and rederive $\mathbb{P}$(triangle). Also find $\mathbb{P}(a > 0.3)$.
Solution. The first piece is $a = \min(U_1, U_2)$ (the leftmost cut). Both cuts must exceed t:

$$\mathbb{P}(a > t) = \mathbb{P}(U_1 > t)\,\mathbb{P}(U_2 > t) = (1-t)^2$$

By symmetry, every piece has this same law. For $t = \frac{1}{2}$: $\mathbb{P}(a > \frac{1}{2}) = \frac{1}{4}$. Two pieces cannot both exceed $\frac{1}{2}$ (they would total more than the stick), so the events $\{a > \frac{1}{2}\}, \{b > \frac{1}{2}\}, \{c > \frac{1}{2}\}$ are mutually exclusive:

$$\mathbb{P}(\text{some bully}) = 3 \cdot \tfrac{1}{4} = \tfrac{3}{4} \;\Rightarrow\; \mathbb{P}(\text{triangle}) = \boxed{\tfrac{1}{4}}$$

Also $\mathbb{P}(a > 0.3) = 0.7^2 = \boxed{0.49}$.
▶ *Speed trick:* "complement + disjoint bullies" beats area-chasing: $1 - 3(1-\frac{1}{2})^2 = \frac{1}{4}$ in one line. The pieces (a,b,c) are exchangeable and uniform on the simplex $a+b+c=1$.

Example 13 — Expected Longest, Middle and Shortest Piece

Find the expected lengths of the longest, middle and shortest pieces of the 3-piece stick.
Solution. For t small enough, the same cut-counting argument gives

$$\mathbb{P}(a > t, b > t) = (1-2t)^2, \qquad \mathbb{P}(a, b, c > t) = (1-3t)^2$$

(valid for $t < \frac{1}{2}$ and $t < \frac{1}{3}$ respectively, else 0).
Shortest: $\mathbb{P}(\min > t) = (1-3t)^2$, so

$$\mathbb{E}[\min] = \int_0^{1/3} (1-3t)^2 dt = \boxed{\tfrac{1}{9}}$$

Longest: by inclusion–exclusion over which pieces exceed t,

$$\mathbb{P}(\max > t) = 3(1-t)^2 - 3(1-2t)_+^2 + (1-3t)_+^2$$

$$\mathbb{E}[\max] = 3 \cdot \tfrac{1}{3} - 3 \cdot \tfrac{1}{6} + \tfrac{1}{9} = \boxed{\tfrac{11}{18}}$$

Middle: the three means sum to 1:

$$\mathbb{E}[\text{mid}] = 1 - \tfrac{11}{18} - \tfrac{1}{9} = \boxed{\tfrac{5}{18}}$$

⋆ *Analogy:* the pattern $\frac{1}{3}\left(\frac{1}{3}\right),\ \frac{1}{3}\left(\frac{1}{2}+\frac{1}{3}\right),\ \frac{1}{3}\left(1+\frac{1}{2}+\frac{1}{3}\right)$ is a staircase: each step up in rank adds another "$\frac{1}{k}$ share". The longest piece averages over half the stick.

Example 14 — Break the Longer Piece Again

Break a unit stick at a uniform point. Then break the *longer* piece at a uniform point. Find $\mathbb{P}$(triangle).
Solution. Let the first cut leave a short piece $s \in (0, \frac{1}{2})$ (density 2 on that range) and a long piece $L = 1-s$. Break L at $w \sim \text{Unif}(0, L)$; the pieces are $s,\ w,\ L-w$.
Need all $< \frac{1}{2}$: $s < \frac{1}{2}$ already holds, and

$$w < \tfrac{1}{2} \;\text{ and }\; L - w < \tfrac{1}{2} \iff w \in \left(L - \tfrac{1}{2},\ \tfrac{1}{2}\right)$$

This window has length $\frac{1}{2} - (L - \frac{1}{2}) = 1 - L = s$, so given s:

$$\mathbb{P}(\text{triangle} \mid s) = \frac{s}{L} = \frac{s}{1-s}$$

$$\mathbb{P} = \int_0^{1/2} 2 \cdot \frac{s}{1-s}\, ds = 2\Big[-s - \ln(1-s)\Big]_0^{1/2} = \boxed{2\ln 2 - 1 \approx 0.386}$$

▶ *Contrast (all verified):* choosing the piece *by length* equals two independent cuts, giving $\frac{1}{4}$; choosing a piece with equal probability gives only $\ln 2 - \frac{1}{2} \approx 0.193$; always breaking the longer piece gives ≈ 0.386. *Always read how the second piece is chosen.*

Example 15 — Every Piece Must Be at Least t

A 1-metre rod is cut into 3 pieces at two uniform points. Find $\mathbb{P}$(every piece ≥ 10 cm) and $\mathbb{P}$(shortest < 5 cm).
Solution. From Example 13, $\mathbb{P}(\text{all pieces} > t) = (1-3t)^2$ for $t \le \frac{1}{3}$. Why: imagine reserving a minimum of t for each piece up front; that uses $3t$ of the stick, leaving a free stick of length $1-3t$ to split arbitrarily — the probability scales as the squared length of what remains.

With $t = 0.1$:

$$\mathbb{P}(\text{all} \geq 0.1) = (1 - 0.3)^2 = \boxed{0.49}$$

With $t = 0.05$:

$$\mathbb{P}(\text{shortest} < 0.05) = 1 - (1 - 0.15)^2 = 1 - 0.7225 = \boxed{0.2775}$$

⋆ *Analogy:* a minimum-order-size rule. Pay the "minimum lot" t to every piece first; whatever stick is left is the only *free* part, and probability is the free area as a fraction of the original area.

D. Capstones — 2D Integrals in Action

Example 16 — Buffon's Needle: Where π Comes From

A needle of length L is dropped on a floor ruled with parallel lines a distance $d \geq L$ apart. Find $\mathbb{P}$(needle crosses a line).
Solution. Let $x \in [0, \frac{d}{2}]$ be the distance from the needle's centre to the nearest line and $\theta \in [0, \frac{\pi}{2}]$ its acute angle to the lines; (x, θ) is uniform on a rectangle of area $\frac{d}{2} \cdot \frac{\pi}{2} = \frac{\pi d}{4}$.
The needle crosses iff its half-height reaches the line: $x \leq \frac{L}{2} \sin\theta$. The favourable area is

$$\int_0^{\pi/2} \tfrac{L}{2} \sin\theta \, d\theta = \tfrac{L}{2}$$

$$\mathbb{P}(\text{cross}) = \frac{L/2}{\pi d/4} = \boxed{\frac{2L}{\pi d}}$$

Check: $L = 1$, $d = 2$ gives $\frac{1}{\pi} \approx 0.318$.
⋆ *Analogy:* the sine curve is the "footprint" of the needle: the more it tilts perpendicular to the lines, the further it reaches. Averaging the footprint over all angles produces the number $\frac{2}{\pi}$ — which is why π appears in a problem with no circle in it.

Example 17 — Does a Random Quadratic Have Real Roots?

The coefficients B, C are iid Uniform$(0, 1)$. Find $\mathbb{P}\big(x^2 + Bx + C = 0$ has real roots$\big)$. Then redo with $B \sim \text{Unif}(0, 2)$, $C \sim \text{Unif}(0, 1)$.
Solution. Real roots iff the discriminant $B^2 - 4C \geq 0$, i.e. $C \leq \frac{B^2}{4}$: the region under a parabola inside the unit square.

$$\text{Area} = \int_0^1 \frac{b^2}{4}\, db = \frac{1}{12} \;\Rightarrow\; \mathbb{P} = \boxed{\tfrac{1}{12}}$$

Variant: the rectangle $[0, 2] \times [0, 1]$ has area 2, and $\frac{b^2}{4} \leq 1$ throughout (since $b \leq 2$), so

$$\text{Area} = \int_0^2 \frac{b^2}{4}\, db = \frac{2}{3} \;\Rightarrow\; \mathbb{P} = \frac{2/3}{2} = \boxed{\tfrac{1}{3}}$$

▶ *Speed trick:* a discriminant condition is a curve in coefficient space. Check the curve stays under the top edge of the rectangle (here $\frac{b^2}{4} \leq 1$ for $b \leq 2$) — if it did not, you would need to clip and split the integral.

Example 18 — A Trading Application: Gains From Trade

A buyer's value V_b and a seller's cost V_s are iid Uniform$(0, 1)$. A trade occurs iff $V_b \geq V_s$. Find $\mathbb{P}$(trade), $\mathbb{E}$[gain | trade] with gain $G = V_b - V_s$, $\mathbb{E}[G\,\mathbf{1}_{\text{trade}}]$, and $\mathbb{E}$[price | trade] if the price is the midpoint.
Solution. $\mathbb{P}(V_b \geq V_s) = \frac{1}{2}$ (the half of the square below the diagonal).
Given a trade, $G = V_b - V_s$ has the same law as $|V_b - V_s|$ (fold the square along the diagonal), so by Example 7 its density is $2(1 - g)$:

$$\mathbb{E}[G \mid \text{trade}] = \int_0^1 g \cdot 2(1 - g)\, dg = \boxed{\tfrac{1}{3}}$$

$$\mathbb{E}[G\,\mathbf{1}_{\text{trade}}] = \tfrac{1}{2} \cdot \tfrac{1}{3} = \boxed{\tfrac{1}{6}}$$

The map $(v_b, v_s) \mapsto (1 - v_s,\, 1 - v_b)$ preserves the trade region and sends the midpoint m to $1 - m$, hence

$$\mathbb{E}[\text{price} \mid \text{trade}] = \boxed{\tfrac{1}{2}}$$

⋆ *Analogy:* half of all potential deals are impossible (the seller wants more than the buyer will pay). In the deals that *do* happen, the surplus is on average only a third of the price range — the rest of the intuition behind why spreads in thin markets matter.

Example 19 — Three Uniforms: the 3D Version

X, Y, Z iid Uniform$(0, 1)$, $S = X + Y + Z$. Find $\mathbb{P}(S \leq 1)$, $\mathbb{P}(S \leq 1.5)$, $\mathbb{P}(S \leq 2)$ and $\mathbb{P}(X < Y < Z)$.
Solution. (X, Y, Z) is uniform on the unit cube (volume 1).
$S \leq 1$ is the corner tetrahedron with legs 1: volume $\frac{1}{3} \cdot \frac{1}{2} \cdot 1 = \boxed{\frac{1}{6}}$.
The map $(x, y, z) \mapsto (1 - x, 1 - y, 1 - z)$ sends $S \to 3 - S$ and preserves the cube, so S and $3 - S$ have the same law:

$$\mathbb{P}(S \leq 1.5) = \boxed{\tfrac{1}{2}}, \qquad \mathbb{P}(S \leq 2) = 1 - \mathbb{P}(S < 1) = \boxed{\tfrac{5}{6}}$$

Finally, all $3! = 6$ orderings of (X, Y, Z) are equally likely:

$$\mathbb{P}(X < Y < Z) = \boxed{\tfrac{1}{6}}$$

⋆ *Analogy:* in 1D the corner is a length (s), in 2D a triangle $(\frac{s^2}{2})$, in 3D a tetrahedron $(\frac{s^3}{6})$: the "simplex volume" is $\frac{s^n}{n!}$ — the same $\frac{1}{n!}$ that counts orderings.

Example 20 — Capstone: Is the Triangle Acute? $(3\ln 2 - 2)$

Break a unit stick at two uniform points. Find $\mathbb{P}$(the pieces form an *acute* triangle).
Solution. Pieces (a, b, c) are uniform on $a + b + c = 1$; in the (a, b)-plane this is $\{a, b > 0,\ a + b < 1\}$ with density 2. A triangle exists iff all pieces are $< \frac{1}{2}$ (probability $\frac{1}{4}$, Example 11).
Obtuse at c: $c^2 > a^2 + b^2$ with $c = 1 - a - b$. Expanding,

$$(1 - a - b)^2 > a^2 + b^2 \iff 1 - 2a - 2b + 2ab > 0 \iff b < \frac{1 - 2a}{2(1 - a)}$$

If also $c < \frac{1}{2}$ (i.e. $b > \frac{1}{2} - a$), then $c > a, b$ automatically and the triangle exists. So the obtuse-at-c region is, for $a \in (0, \frac{1}{2})$,

$$\tfrac{1}{2} - a \;<\; b \;<\; \tfrac{1-2a}{2(1-a)}$$

$$\text{Width} = (1 - 2a)\Big(\tfrac{1}{2(1-a)} - \tfrac{1}{2}\Big) = \frac{a(1 - 2a)}{2(1 - a)}$$

Divide: $\frac{a(1-2a)}{1-a} = 2a + 1 - \frac{1}{1-a}$, so

$$\text{Area} = \tfrac{1}{2} \int_0^{1/2} \Big(2a + 1 - \tfrac{1}{1-a}\Big) da = \tfrac{1}{2}\big(\tfrac{3}{4} - \ln 2\big)$$

Density 2 gives $\mathbb{P}$(obtuse at c, triangle) $= \frac{3}{4} - \ln 2$. Three symmetric (disjoint) choices of the obtuse side:

$$\mathbb{P}(\text{acute}) = \tfrac{1}{4} - 3\Big(\tfrac{3}{4} - \ln 2\Big) = \boxed{3\ln 2 - 2 \approx 0.0794}$$

Hence, given that a triangle forms, $\mathbb{P}(\text{acute} \mid \text{triangle}) \approx 0.318$ and $\mathbb{P}(\text{obtuse} \mid \text{triangle}) \approx 0.682$.
▶ *Full-circle intuition:* the same two-cut simplex powers every part of this chapter — area ratios (Ex. 1), a clean polygon (Ex. 11), and finally a curved boundary (Ex. 10, 17) needing a logarithm. A random triangle from a broken stick is overwhelmingly *flat*: one piece nearly the length of the other two together.